LMS Line-Following Robot — Bill of Materials

This PDF reflects the current values in BOM_LMS_robot.xlsx.

1. Bill of Materials

No.	Item (exact model)	Qty	Unit price (INR)	Line total (INR)	Supplier	Product / search URL	Notes
1	ESP32 DevKit V1 (ESP32-WROOM-32, 30-pin, CP2102)	2	Rs 499.00	Rs 998.00	Probots	https://probots.co.in/esp32-30-pin-cp2102-development-board.html	Controller. 1 spare: GPIO damage is the most common failure. Listed at Rs 499 incl. GST.
2	TCRT5000 Reflective IR Sensor Photoelectric Switch Pack of 5	2	Rs 69.00	Rs 138.00	Robu.in	https://robu.in/product/tcrt5000-reflective-ir-sensor-photoelectric-switch/	5 for array + 3 spare. Use AO (analog), run from 3.3 V. Price not shown in search snippet; confirm on page.
3	N20 12 V 300 RPM micro metal gear motor WITH encoder	1	Rs 514.48	Rs 514.48	Zbotic	https://zbotic.in/product/n20-12v-300rpm-micro-metal-gear-motor-with-encoder/	Upgrade over the plain motor: encoders give speed feedback. Rs 436 + GST.
4	N20 motor bracket (pair-compatible)	5	Rs 27.80	Rs 139.00	Amazon	https://www.amazon.in/Motor-Mount-Bracket-N20-2pcs/dp/B094ZJVZ1L	2 in use + 2 spare. Search link: pick the bracket sold for N20.
5	43 mm rubber-tyre wheel for N20 (3 mm D-shaft)	2	Rs 38.00	Rs 76.00	Robu.in	https://robu.in/?s=43mm+wheel+N20&post_type=product	~0.52 m/s no-load at ~9.3 V.
6	Metal ball caster 3/8 in	2	Rs 80.00	Rs 160.00	Probots	https://probots.co.in/catalogsearch/result/?q=metal+ball+caster	Low-friction 3rd contact point; 1 spare.
7	L298N dual H-bridge motor driver module	2	Rs 149.00	Rs 298.00	Probots	https://probots.co.in/l298n-2a-dual-motor-driver-module.html	Matches Appendix A pin map. 1 spare (easy to burn during bring-up). Rs 149 incl. GST.
8	CNHL MiniStar 3S 11.1 V 1000 mAh 100C LiPo, XT60	2	Rs 1,066.72	Rs 2,133.44	Zbotic	https://zbotic.in/product/cnhl-ministar-1000mah-11-1v-3s-100c-lipo-battery-with-xt60-plug/	Two packs: run one while the other charges. Rs 904 + GST each, free shipping stated.
9	IMAX RC B3 Pro balance charger 2S/3S (EU plug)	1	Rs 499.00	Rs 499.00	Probots	https://probots.co.in/imax-b3-pro-balance-charger-for-2s-3s-lipo-battery.html	Mandatory for safe LiPo charging. Rs 499 incl. GST.
10	LiPo low-voltage buzzer alarm (1-8S)	2	Rs 150.00	Rs 300.00	Robu.in	https://robu.in/?s=lipo+low+voltage+buzzer+alarm&post_type=product	One per pack. Stops over-discharge below ~3.3 V/cell.
11	XT60 connector pair (robot side)	2	Rs 60.00	Rs 120.00	Probots	https://probots.co.in/catalogsearch/result/?q=XT60+connector	Robot-side battery connector; 1 spare.
12	Rocker switch 6 A	2	Rs 35.00	Rs 70.00	Robu.in	https://robu.in/?s=rocker+switch+6A&post_type=product	Main power switch.
13	Blade fuse holder + 5 A fuse	2	Rs 80.00	Rs 160.00	Robu.in	https://robu.in/?s=blade+fuse+holder&post_type=product	Short-circuit protection (worst case 1.9 A).
14	MP1584EN buck module (set to 5.0 V)	2	Rs 79.00	Rs 158.00	Probots	https://probots.co.in/mp1584-dc-dc-step-down-3a-adjustable-buck-module.html	5 V rail for ESP32 Vin + L298N logic. 1 spare: the trimpot is fragile. Rs 79 incl. GST.
15	3 mm acrylic chassis plate, laser cut (or 3D-print)	1	Rs 450.00	Rs 450.00	Local fabricator	https://robu.in/?s=robot+chassis+acrylic&post_type=product	Custom 2-deck chassis; local laser-cut price is a guess. Search link is only a fallback.
16	M3 standoff + screw kit	1	Rs 200.00	Rs 200.00	Probots	https://probots.co.in/catalogsearch/result/?q=M3+standoff+screw+kit	Mounting decks.
17	Perfboard 5x7 cm	4	Rs 30.00	Rs 120.00	Probots	https://probots.co.in/catalogsearch/result/?q=perfboard+5x7	Sensor-array / power distribution boards.
18	Jumper wire set M-F + F-F 20 cm (40 pcs)	2	Rs 80.00	Rs 160.00	Robu.in	https://robu.in/?s=jumper+wire+40+pcs&post_type=product	Prototyping connections.
19	22 AWG silicone wire 2 m (red/black)	1	Rs 150.00	Rs 150.00	Robu.in	https://robu.in/?s=22+AWG+silicone+wire&post_type=product	Power wiring.
20	Header strips + 2-pin screw terminals	1	Rs 120.00	Rs 120.00	Probots	https://probots.co.in/catalogsearch/result/?q=header+strip+screw+terminal	Connectors.
21	Capacitor + resistor kit (1000 uF/25 V, 100 nF, 10 uF, 10 k)	1	Rs 150.00	Rs 150.00	Robu.in	https://robu.in/?s=capacitor+kit+1000uF&post_type=product	Motor-rail bulk cap, decoupling, and the 10 k pull-down for GPIO12.
22	Heat-shrink, zip ties, double-sided tape	1	Rs 120.00	Rs 120.00	Robu.in	https://robu.in/?s=heat+shrink+tube+assortment&post_type=product	Assembly.

No.	Item (exact model)	Qty	Unit price (INR)	Line total (INR)	Supplier	Product / search URL	Notes
23	Micro-USB data cable	1	Rs 100.00	Rs 100.00	Probots	https://probots.co.in/catalogsearch/result/?q=micro+usb+data+cable	Programming (must be a data cable).
24	Velcro battery strap	1	Rs 50.00	Rs 50.00	Robu.in	https://robu.in/?s=velcro+battery+strap&post_type=product	Battery retention.
25	Digital multimeter	1	Rs 400.00	Rs 400.00	Robu.in	https://robu.in/?s=digital+multimeter&post_type=product	Needed to set the buck to 5.0 V BEFORE connecting the ESP32, and to check pack voltage.
26	LiPo safe charging bag	1	Rs 250.00	Rs 250.00	Robu.in	https://robu.in/?s=lipo+safe+bag&post_type=product	Fire-resistant bag for charging/storage.

2. Budget Summary

LMS Line-Following Robot: Budget Summary	
Maximum budget	Rs 10,000.00
Parts subtotal	Rs 7,680.92
Shipping (assumed, 3 orders)	Rs 400.00
Committed spend	Rs 8,080.92
Reserve / contingency (budget left)	Rs 1,919.08
TOTAL (committed + reserve)	Rs 10,000.00
Reserve as % of budget	19.19%
Status	Within budget
Spend by supplier (parts only)	
Probots	2773
Zbotic	2647.92
Robu.in	1810
Local fabricator	450
Share of parts cost with a verified listing price	#REF!
Legend	
Blue numbers = inputs you can change (budget, shipping, qty, unit price).	
Green 'Price basis' = price read from a live shop listing. Yellow = my estimate: check it before ordering.	
Black = formulas; everything recalculates when you edit qty or price.	

3. Notes from the Excel workbook

• Verified prices came from product pages found by web search while preparing this sheet (late Oct 2026). Shop prices change daily: re-check each before paying.
• Verified items: ESP32 (Probots), L298N (Probots), IMAX B3 Pro charger (Probots), MP1584 buck (Probots), CNHL MiniStar 3S LiPo (Zbotic), N20 12V 300RPM encoder motor (Zbotic).
• Estimated items: I could not confirm a live price for the rest. Their URLs are store search links, not individual product pages. Open the link, pick the matching part, and update the blue unit price.
• The N20 encoder motor page was marked out of stock on a Zbotic category listing: confirm before ordering. The encoder pins are not used by the current firmware; they are for future speed feedback.
• Shipping of Rs 400 is an assumption for 3 supplier orders; Zbotic states free shipping on orders above Rs 999.
• Changes from the original BOM: 2 batteries, spare ESP32 / L298N / buck / sensors, encoder motors, multimeter and LiPo bag added to use the full Rs 10,000 budget safely.
• Safety: charge LiPo only with the balance charger, in the safe bag, and never unattended. Set the buck to 5.0 V with the multimeter before connecting the ESP32.

Source workbook: BOM_LMS_robot.xlsx. Prices and assumptions shown here are those currently stored in the workbook; the PDF does not independently verify live shop prices.